CS 255 Business Requirements Document

Yunus Parvez Khan
(yunus.parvezkhan@snhu.edu)

Department of Computer Science
Course Title: System Analysis and Design (CS 255)
Instructor: Dr. Anamika Singh
Southern New Hampshire University
Manchester, NH 03106, United States

System Components and Design

Purpose

What is the purpose of this project? Who is the client and what do they want their system to be able to do?

- The purpose of this project is to design a centralized, web-based system for DriverPass, a driver-training business established by Liam to address the need for better preparation among students taking their driving license examinations.
- The system shall integrate online classes, practice tests, in-person instruction on DMV rules and policies, and on-the-road driving lessons into a unified platform, according to the training package selected by each customer.
- The system shall support the management of customer accounts, training packages, driving lesson reservations, driver and vehicle assignments, and student training records.
- The system shall provide customers with online access to their accounts and allow them to schedule, modify, and cancel driving lessons, while enabling the secretary to manage appointments for customers who contact or visit the office.
- The system shall support the operational needs of the owner, secretary, and IT officer through role-appropriate access to business information, account management functions, activity tracking, and reporting.
- The system shall be accessible through web-enabled computers and mobile devices, preferably using a cloud-based environment, and shall support the secure handling of information and the downloading of reports for external use.
- Ultimately, the project aims to provide DriverPass with an integrated system that improves the organization and accessibility of its training services while helping students prepare for both the theoretical and practical aspects of their driving examinations.

System Background

What does DriverPass want the system to do? What is the problem they want to fix? What are the different components needed for this system?

- DriverPass is a driver-training business established by Liam to address the lack of comprehensive preparation available to students taking their local Department of Motor Vehicles (DMV) driving

examinations. The business intends to combine theoretical instruction with practical driving experience through a single service platform.

- The system shall provide three initial training packages. Package One includes six hours of on-the-road instruction; Package Two includes eight hours of on-the-road instruction and an in-person lesson covering DMV rules and policies; and Package Three includes twelve hours of on-the-road instruction, the in-person lesson, and access to online course materials and practice tests.
- The system shall support the scheduling and management of two-hour driving lessons. Each reservation must be associated with the relevant customer, scheduled date and time, assigned driver, and vehicle. Customers shall be able to manage their appointments online, while the secretary shall be able to schedule appointments on their behalf.
- The system shall maintain customer information, training-package registrations, driving lesson records, and online test progress. Test records shall include the test name, time taken, score, and status, which may be not taken, in progress, failed, or passed. Driver notes and lesson times shall also be available for reviewing student training history.
- DriverPass requires different levels of access for its business users. The owner needs access to business information and activity reports, the secretary needs customer-registration and appointment-management functions, and the IT officer needs system-maintenance and account-management capabilities, including resetting passwords and blocking access when necessary.
- The system shall maintain an activity history identifying who created, modified, or canceled reservations. This information shall support accountability by allowing authorized users to review and print activity reports when investigating scheduling issues.
- The system shall operate through the web, preferably in a cloud-based environment, and allow authorized users to access information from different computers and mobile devices. It shall also support downloading reports and selected business information for external use, such as analysis in Microsoft Excel.
- The system shall support DriverPass's need to keep its educational materials current with DMV requirements. The client expects a connection through which updated rules, policies, and sample questions can be received, together with notifications when updates occur.
- The system must accommodate changes in DriverPass's operational needs without requiring a complete redesign. In particular, the owner must be able to disable a training package when new registrations for that package are no longer desired. Adding or removing entire software modules is outside the scope of the initial release and may be considered for future development.

Objectives and Goals

What should this system be able to do when it is completed? What measurable tasks need to be included in the system design to achieve this?

- The primary objective of the DriverPass system is to provide a centralized platform for managing driver-training services, customer information, and business operations while allowing customers to access their training resources and appointments online.
- The system shall maintain customer registration records containing the customer's first name, last name, address, phone number, state, credit-card information, pickup location, and drop-off location.
- The system shall allow customers to register for available training packages and access the services included in their selected packages.

- The system shall support the scheduling of two-hour driving lessons and maintain reservation records that identify the customer, lesson date and time, assigned driver, and assigned vehicle.
- The system shall allow customers to create, modify, and cancel driving lesson reservations through their accounts, while enabling the secretary to schedule appointments for customers who call or visit the office.
- The system shall maintain records of the three initial training packages and allow the owner to disable a package when DriverPass no longer wishes to accept new registrations for it.
- The system shall provide online course materials and practice tests to customers whose selected training package includes these services.
- The system shall record each customer's practice-test progress, including the test name, time taken, score, and status, with status values of not taken, in progress, failed, or passed.
- The system shall maintain driving lesson records that include lesson times and comments provided by the assigned driver, allowing authorized users to review a student's training history.
- The system shall enforce role-based access permissions for customers, the owner, the secretary, and the IT officer according to their responsibilities within DriverPass.
- The system shall allow the IT officer to manage user accounts, reset forgotten passwords, and block access when necessary.
- The system shall record reservation creation, modification, and cancellation activities, identifying the user responsible for each change, and allow authorized users to generate and print activity reports.
- The system shall allow the owner to access business information and download reports or selected data for external use, including analysis in Microsoft Excel.
- The system shall provide authorized access to system data through web-enabled computers and mobile devices, preferably using a cloud-based environment.
- The system shall support the receipt of DMV updates concerning driving rules, policies, and sample questions and notify DriverPass when updates become available.
- The system shall provide appropriate interfaces for customer registration, appointment management, student test progress, driver notes, and communication between DriverPass and its customers.

Requirements

Nonfunctional Requirements

In this section, you will detail the different nonfunctional requirements for the DriverPass system. You will need to think about the different things that the system needs to function properly.

Performance Requirements

What environments (web-based, application, etc.) does this system need to run in? How fast should the system run? How often should the system be updated?

- The system shall be accessible through a web-based interface that can be used from computers and mobile devices with an internet connection.
- The system shall provide timely responses to user requests, including account access, customer information retrieval, appointment scheduling, and practice-test progress viewing, under normal operating conditions.

- The system shall support centralized data storage so that authorized users can access consistent and up-to-date information across supported devices.
- The system shall allow authorized users to access existing records when working remotely, including from locations outside the DriverPass office. Changes to system records shall require an active connection to avoid conflicting or duplicate data.
- The system shall support the downloading of business reports and selected records in a format suitable for use with external applications such as Microsoft Excel.
- The system shall support routine software maintenance and updates to keep the platform operational and compatible with supported web browsers and devices.
- The system shall operate in a cloud-based environment, preferably managed by a service provider, to reduce DriverPass's responsibility for maintaining physical infrastructure, backups, and security.
- The system shall support the storage and retrieval of customer information, training-package records, reservations, driver assignments, vehicle assignments, and practice-test results without unintended data loss or duplication.
- The system shall preserve the accuracy of customer records and reservation details by validating required fields and checking the consistency of dates, times, and assigned resources before saving changes.
- The system shall maintain accurate associations between customers, training packages, lesson reservations, drivers, and vehicles so that authorized users can retrieve the correct information when needed.

Platform Constraints

What platforms (Windows, Unix, etc.) should the system run on? Does the back end require any tools, such as a database, to support this application?

- The system shall support unique identification of each user account to ensure that customer and employee records are associated with the correct individual.
- The system shall validate user-provided information, including required customer registration fields, contact details, reservation dates, and appointment times, before accepting or updating records.
- The system shall distinguish between valid and invalid reservation requests and prevent the assignment of a driver or vehicle when the requested resource is already committed to another lesson at the same time.
- The system shall maintain consistent status values for practice tests, using the categories specified by DriverPass: not taken, in progress, failed, and passed.
- The system shall record the identity of the user responsible for each reservation creation, modification, or cancellation and preserve the associated activity history for accountability.
- The system shall notify the appropriate authorized personnel when a system error or operational issue prevents a critical function, such as reservation processing or record updating, from completing successfully.
- The system shall provide clear feedback when user input is incomplete, invalid, or inconsistent, allowing the user to correct the information before resubmitting it.
- The system shall ensure that customer records, lesson reservations, and training-package registrations remain associated with the correct accounts when records are retrieved, modified, or reported.

- The system shall apply consistent validation rules to relevant inputs regardless of whether an appointment or customer record is entered by a customer through the website or by the secretary through the office interface.

Accuracy and Precision

How will you distinguish between different users? Is the input case-sensitive? When should the system inform the admin of a problem?

- The system shall allow authorized personnel to add, modify, deactivate, and manage user accounts according to their assigned responsibilities and access permissions.
- The system shall allow the IT officer to reset user passwords and block account access when an employee leaves the organization or when access must otherwise be revoked.
- The system shall support changes to customer information and other operational records through authorized interfaces without requiring modifications to the underlying application code for routine record management.
- The system shall allow the owner to disable a training package when new customer registrations for that package are no longer desired, while retaining the information needed to manage existing registrations appropriately.
- The system shall support routine software and platform updates while preserving stored customer records, reservation histories, training-package information, and practice-test results.
- The system shall allow the IT officer to perform necessary system-maintenance activities while restricting access to administrative functions from unauthorized users.
- The system shall accommodate the addition or modification of ordinary business records as DriverPass's operational needs change, without requiring a complete redesign of the system.
- The system shall maintain a distinction between configurable business information, such as training-package availability, and changes to software modules that require developer or systems analyst involvement.

Adaptability

Can you make changes to the user (add/remove/modify) without changing code? How will the system adapt to platform updates? What type of access does the IT admin need?

- The system shall allow authorized users to add, modify, and deactivate user accounts and update customer information without requiring changes to the application code for routine account management.
- The system shall allow the owner to disable a training package when DriverPass no longer wishes to accept new registrations for it, without requiring a software update.
- The system shall maintain flexibility in managing training-package availability while preserving existing customer registrations and associated training records.
- The system shall allow the IT officer to reset passwords, manage user accounts, and revoke access when necessary, supporting changes in employee responsibilities and staffing.
- The system shall support routine software updates and compatibility adjustments to accommodate changes in supported web browsers, computers, and mobile devices while preserving existing business data.

- The system shall allow authorized users to access the system from different supported devices without requiring separate customer or employee records for each device.
- The system shall support future expansion of business records and routine operational data as DriverPass grows, within the capabilities of the implemented system.
- The system shall distinguish between routine configuration changes, such as disabling a training package, and adding or removing software modules, which will require developer or systems analyst involvement and may be considered in a future release.

Security

What is required for the user to log in? How can you secure the connection or the data exchange between the client and the server? What should happen to the account if there is a "brute force" hacking attempt? What happens if the user forgets their password?

- The system shall require users to authenticate with valid account credentials before accessing protected information or performing account-specific operations.
- The system shall enforce role-based access control to ensure that customers, the owner, the secretary, and the IT officer can access only the functions and information permitted by their assigned responsibilities.
- The system shall protect customer personal information, contact details, reservation records, training progress, and other sensitive business data from unauthorized access, disclosure, modification, or deletion.
- The system shall protect data transmitted between users' devices and the system through secure communication channels.
- The system shall provide an account-recovery mechanism that allows users to reset forgotten passwords through an appropriate verification process.
- The system shall allow the IT officer to reset user passwords and block or revoke account access when necessary, including when an employee leaves DriverPass or an account is suspected of being compromised.
- The system shall implement safeguards against repeated unauthorized login attempts, including appropriate restrictions or temporary account lockouts when suspicious activity indicates a possible brute-force attack.
- The system shall record significant security-sensitive activities, including reservation creation, modification, and cancellation, with the identity of the responsible user to support accountability and investigation.
- The system shall restrict access to administrative functions, account-management controls, and business reports to authorized personnel.
- The system shall support appropriate backup and recovery arrangements to reduce the risk of permanent data loss following system failures or other operational incidents.
- The system shall handle credit-card information securely and restrict its access to authorized processes and personnel in accordance with applicable security requirements.
- The system shall provide clear error messages for authentication failures without unnecessarily revealing sensitive account information that could assist unauthorized access attempts.

Functional Requirements

Using the information from the scenario, think about the different functions the system needs to provide. Each of your bullets should start with "The system shall . . ." For example, one functional requirement might be, "The system shall validate user credentials when logging in."

- The system shall allow customers to create accounts and maintain their registration information, including first name, last name, address, phone number, state, credit-card information, pickup location, and drop-off location.
- The system shall allow customers to log in to their accounts and reset forgotten passwords through an automated account-recovery process.
- The system shall allow customers to view the available training packages and register for a package that is open for enrollment.
- The system shall provide three initial training packages: Package One with six hours of driving instruction, Package Two with eight hours of driving instruction and an in-person lesson on DMV rules and policies, and Package Three with twelve hours of driving instruction, the in-person lesson, and online classes with practice tests.
- The system shall allow the owner to disable a training package so that new customers cannot register for it when DriverPass no longer wishes to offer it.
- The system shall allow customers to schedule two-hour driving lessons by selecting their preferred day and time.
- The system shall allow customers to modify or cancel their existing driving lesson reservations through their accounts.
- The system shall allow the secretary to register customers and create, modify, or cancel appointments on their behalf when they call or visit the DriverPass office.
- The system shall associate each driving lesson reservation with the appropriate customer, scheduled date and time, assigned driver, and assigned vehicle.
- The system shall maintain records of the company's drivers and vehicles to support the assignment of appropriate resources to scheduled driving lessons.
- The system shall provide online course materials and practice tests to customers whose selected training package includes those services.
- The system shall record practice-test information, including test name, time taken, score, and status, and display whether each test has not been taken, is in progress, has failed, or has passed.
- The system shall maintain driving lesson records, including lesson times and comments entered by the assigned driver.
- The system shall allow the owner to access business information and download reports or selected records for external use, including Microsoft Excel.
- The system shall generate activity reports that identify who created, modified, or canceled a reservation and allow authorized personnel to print these reports.
- The system shall provide the IT officer with account-management functions, including resetting passwords and blocking access when necessary.
- The system shall provide a mechanism for receiving updates from the DMV concerning driving rules, policies, and sample questions.
- The system shall notify the appropriate DriverPass personnel when new DMV updates become available.
- The system shall provide interfaces for customer registration, appointment management, practice-test progress, driver notes, and communication between DriverPass and its customers.
- The system shall allow authorized users to access business data through web-enabled computers and mobile devices, including from locations outside the DriverPass office.

- The system shall maintain a record of significant reservation changes so that authorized users can review the history of an appointment when investigating a problem.

User Interface

What are the needs of the interface? Who are the different users for this interface? What will each user need to be able to do through the interface? How will the user interact with the interface (mobile, browser, etc.)?

- The system shall provide a web-based user interface accessible through supported computers and mobile devices, allowing authorized users to access the system from the DriverPass office or other locations with an internet connection.
- The system shall provide a customer registration interface for entering personal information, contact details, credit-card information, and preferred pickup and drop-off locations.
- The system shall provide a customer account interface through which customers can view available training packages, register for a package, and manage their account information.
- The system shall provide an appointment-management interface that allows customers to schedule, modify, and cancel two-hour driving lessons by selecting their preferred dates and times.
- The system shall provide an interface for the secretary to register customers and manage appointments on behalf of customers who call or visit the DriverPass office.
- The system shall provide an interface for viewing driving lesson schedules and the associated customer, assigned driver, lesson time, and vehicle information.
- The system shall provide an online learning interface that allows eligible customers to access course materials and practice tests according to their selected training package.
- The system shall display a practice-test progress section showing the test name, time taken, score, and status, with status values of not taken, in progress, failed, or passed.
- The system shall provide a lesson-record interface displaying lesson times and comments entered by the assigned driver so authorized users can review a student's training history.
- The system shall provide the owner with an interface for accessing business information, reviewing activity reports, downloading selected records, and printing reports when needed.
- The system shall provide the IT officer with an administrative interface for managing user accounts, resetting passwords, blocking access, and performing authorized system-maintenance activities.
- The system shall provide appropriate navigation and access controls so that each user sees the functions and information relevant to their assigned role.
- The system shall provide clear confirmation messages when users successfully create, modify, or cancel appointments and clear error messages when submitted information is incomplete or invalid.
- The system shall provide a communication interface through which customers can contact DriverPass and authorized staff can access the information necessary to respond to customer inquiries.
- The system shall display notifications to appropriate DriverPass personnel when updates to DMV rules, policies, or sample questions are received.
- The system shall present information in a consistent and understandable format, allowing users to identify appointment details, training-package information, and student progress without unnecessary navigation.

Assumptions

What things were not specifically addressed in your design above? What assumptions are you making in your design about the users or the technology they have?

- It is assumed that DriverPass will provide the necessary business information, training materials, DMV-related content, and operational policies required to configure the system.
- It is assumed that customers will have access to a computer or mobile device with an internet connection to use the web-based platform and manage their accounts.
- It is assumed that the system will be hosted in a cloud-based environment, with appropriate arrangements for data storage, backups, maintenance, and security to reduce DriverPass's technical administration responsibilities.
- It is assumed that the owner, secretary, and IT officer will receive appropriate user accounts and permissions based on their respective responsibilities within DriverPass.
- It is assumed that DriverPass will maintain accurate information about its customers, drivers, vehicles, training packages, and available lesson times to support reliable scheduling.
- It is assumed that each driving lesson will last two hours and that the system will use the availability of drivers and vehicles to support the assignment of resources to reservations.
- It is assumed that the three training packages described in the interview will be available in the initial release, subject to the owner's ability to disable a package when new registrations are no longer desired.
- It is assumed that the DMV will provide an accessible method for communicating updates to driving rules, policies, or sample questions. The specific technical method for this connection has not been established in the interview.
- It is assumed that the system will support authorized downloading of business reports and selected records in a format compatible with external applications such as Microsoft Excel.
- It is assumed that the system will maintain centralized records and require an active internet connection for changes to be saved, preventing conflicting updates across disconnected copies of the data.
- It is assumed that DriverPass will establish appropriate procedures for handling customer credit-card information and will ensure that the collection and storage of such information comply with applicable security requirements.
- It is assumed that adding or removing entire software modules will remain outside the initial release, while routine account management and training-package availability changes will be supported by the implemented system.

Limitations

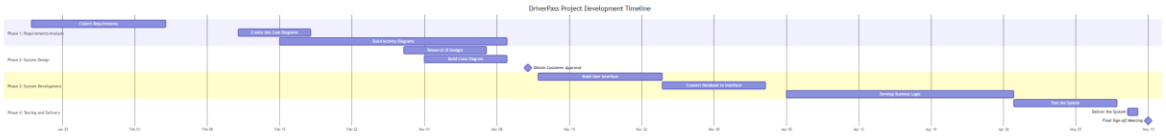
Any system you build will naturally have limitations. What limitations do you see in your system design? What limitations do you have as far as resources, time, budget, or technology?

- The system will require an active internet connection to save changes and update centralized records. Although authorized users may access information remotely, modifying records while offline will not be supported.
- The system's initial release will be limited to the three training packages described in the interview. Adding or removing entire software modules will require developer or systems analyst involvement and is outside the scope of the initial release.

- The system's functionality will depend on the availability and reliability of the cloud-hosting environment, network connectivity, and supporting services.
- The system's ability to keep training materials current with DMV requirements will depend on the availability of an appropriate method for receiving updates from the DMV. The interview does not specify the technical integration method or guarantee the availability of an automated interface.
- The system will support the management of driving lesson reservations, but its scheduling accuracy will depend on DriverPass maintaining up-to-date information about driver availability, vehicle availability, and existing appointments.
- The system will initially provide the three predefined training packages and will not include unrestricted customization of package structures by nontechnical users.
- The system's reporting capabilities will be limited to the information collected and maintained within the system and the report formats implemented during development.
- The system will provide access according to defined user roles and permissions; users will not be able to perform administrative or management functions beyond their assigned access rights.
- The system's ability to provide secure and reliable services will depend partly on appropriate configuration, maintenance, backup arrangements, and security practices within the selected cloud environment.
- The interview does not establish specific performance benchmarks, maximum numbers of concurrent users, availability targets, or recovery-time requirements. These details will need to be determined during further requirements clarification and system design.
- The interview does not define the specific payment-processing method or external payment service, so the initial design cannot assume a particular payment gateway or payment-processing integration without further client confirmation.
- The project schedule and scope are based on the information provided during the initial interview. Significant changes to requirements or additional features requested later may require adjustments to the schedule, resources, or development effort.

Gantt Chart

Please include a screenshot of the GANTT chart that you created with Lucidchart. Be sure to check that it meets the plan described by the characters in the interview.



Link to access the gantt chart through mermaid.live for better understanding: click here

Conclusion

The proposed DriverPass system provides an integrated solution to the business's need for organizing driver training, online learning, customer registration, and driving lesson reservations. By centralizing customer records, training packages, appointment scheduling, driver and vehicle assignments, and student progress information, the system can improve operational efficiency and provide customers with a more accessible learning experience. Role-based access, activity tracking, reporting capabilities, and account management functions also support accountability and the protection of business information.

The requirements analysis establishes a foundation for translating DriverPass's operational needs into functional and nonfunctional system requirements. The proposed user interface and system capabilities are designed to support customers, the secretary, the owner, and the IT officer according to their respective responsibilities. Furthermore, the project schedule organizes the development process into requirements analysis, system design, development, testing, and delivery, while identifying activities that can proceed concurrently. Although implementation details such as the DMV integration method, payment processing mechanism, and specific performance targets require further clarification, the documented requirements provide a structured starting point for system development. Overall, the proposed system aims to help DriverPass deliver a more organized, secure, and scalable driver training service while remaining aligned with its current business objectives.